

Economics 2220

Principles of Macroeconomics

Professor Zhigang Feng

Lecture 1: Introduction

- Course presentation
- Introduction to Macroeconomics
- Recent and Current macroeconomic Events
- List of topics

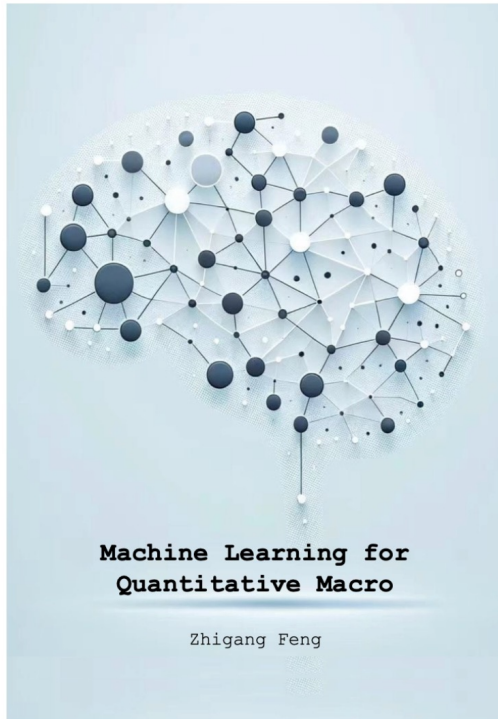
Course Presentation

- Course website: Canvas.
- The choice of textbook.
- Homework, exams and grades.
- Office hours, course materials and additional readings.

Who am I?

- A macroeconomist studies AI and Machine Learning;
- Taught at U of Illinois, U of Zurich, Purdue;
- Conduct researches at Congressional Budget Office, Federal Reserve Bank;
- Published at top economics journals;
- Contributor to FOX 42 KPTM, KETV, Omaha World Herald;
- Photographer for MNHS Swimming Team, 2021-2025

Machine Learning for Quant Macro



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- **Table of Content**, [pdf](#)
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- **Chapter 2:** Neural Networks for Machine Learning, [pdf](#)
- **Chapter 3:** Machine Learning with PyTorch, [pdf](#)
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- **Chapter 5:** Solving Dynamic Equilibrium Model using Machine Learning, [pdf](#)
- **Chapter 6:** Heterogenous Agents Model via Machine Learning, [pdf](#)
- **Chapter 7:** Numerical Simulation of Non-optimal Economies via Reinforcement Learning, [pdf](#)

The Economist's Guide to AI: Fusing Concepts, Context, and Discovery

Zhigang Feng

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HEALTH, HEALTH INSURANCE, AND INEQUALITY

Chaoran Chen, Zhigang Feng , Jiaying Gu

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 SECTIONS

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Abstract

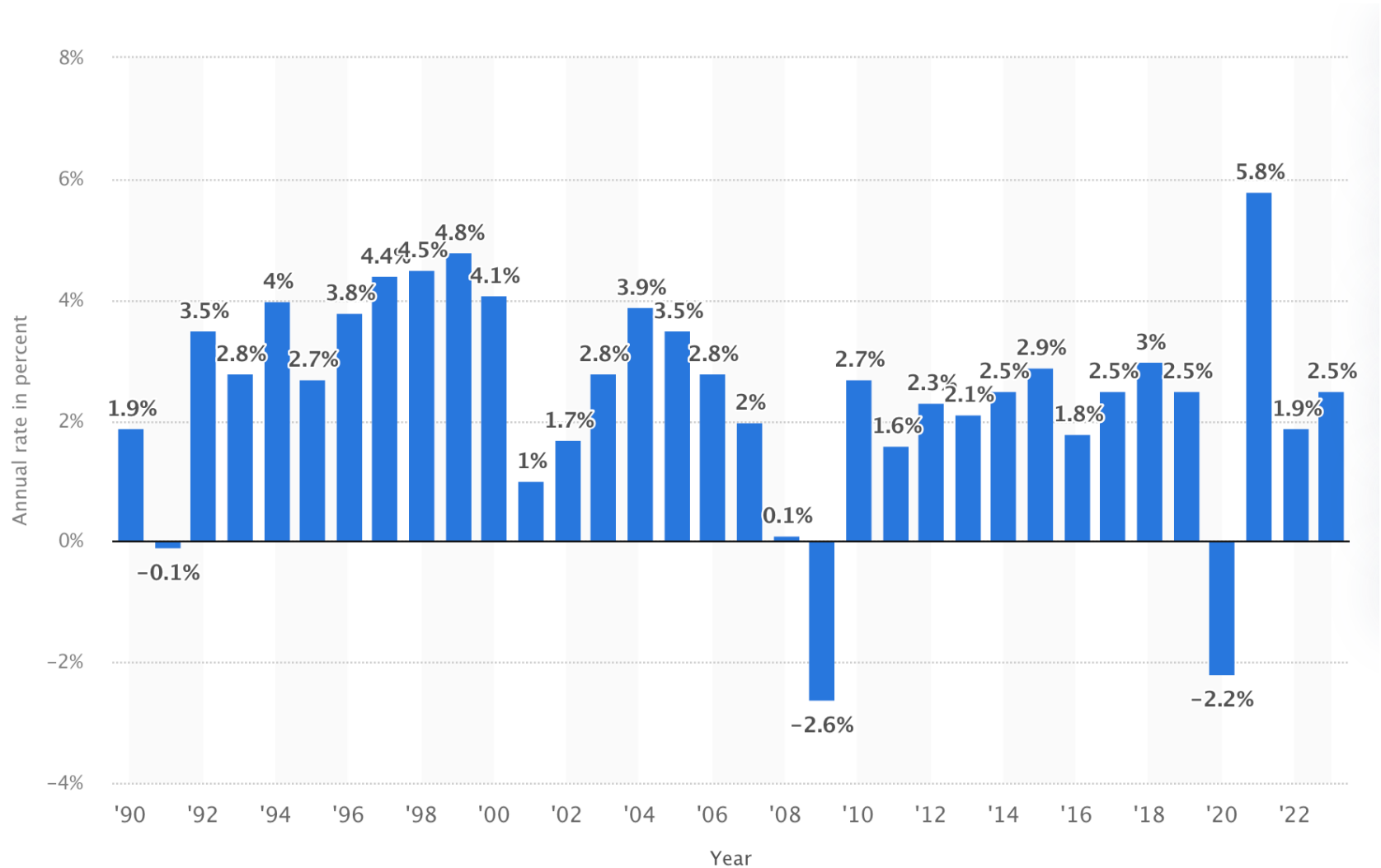
This article identifies a “*health premium*” of insurance coverage: insured individuals are more likely to maintain good health or recover from poor health. We introduce this feature into a prototypical macrohealth model and estimate the baseline economy by matching the observed joint distribution of health insurance, health, and income over the life cycle. Quantitative analysis reveals that an individual's insurance status has a substantial and persistent impact on health. Providing universal health coverage would narrow health and life expectancy gaps, with a mixed effect on the income distribution in the absence of any additional redistribution of income or wealth.

Topaz Vi

<https://www.unomaha.edu/news/2025/10/is-ai-luring-everyone-into-a-middle-intelligence-trap.php>

<https://www.unomaha.edu/news/2025/08/when-hail-meets-high-finance-nebraskas-insurance-crisis-points-to-americas-climate-future.php>

Economic Growth



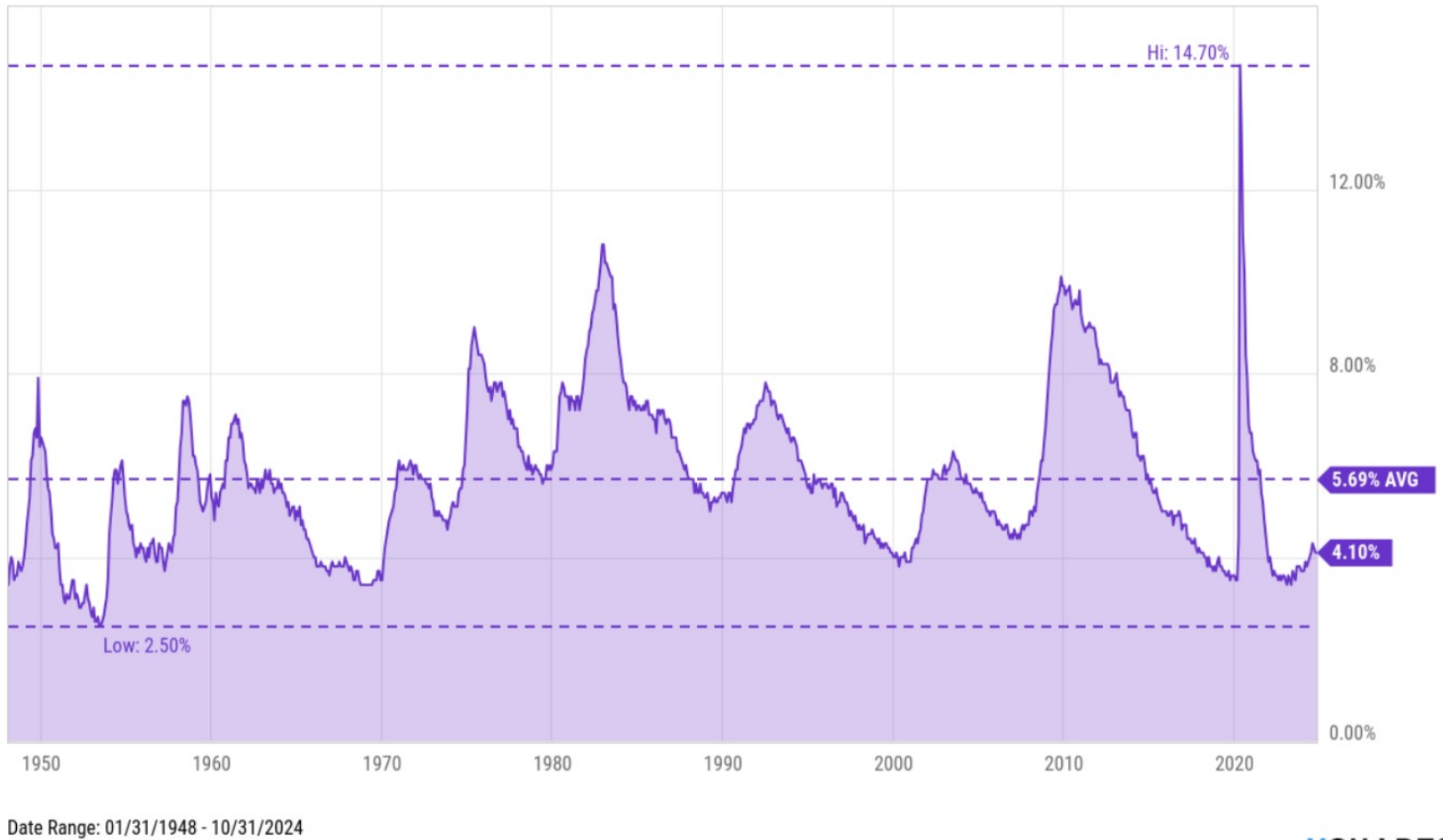
US GDP growth

Real GDP, % change on previous quarter, at annual rates

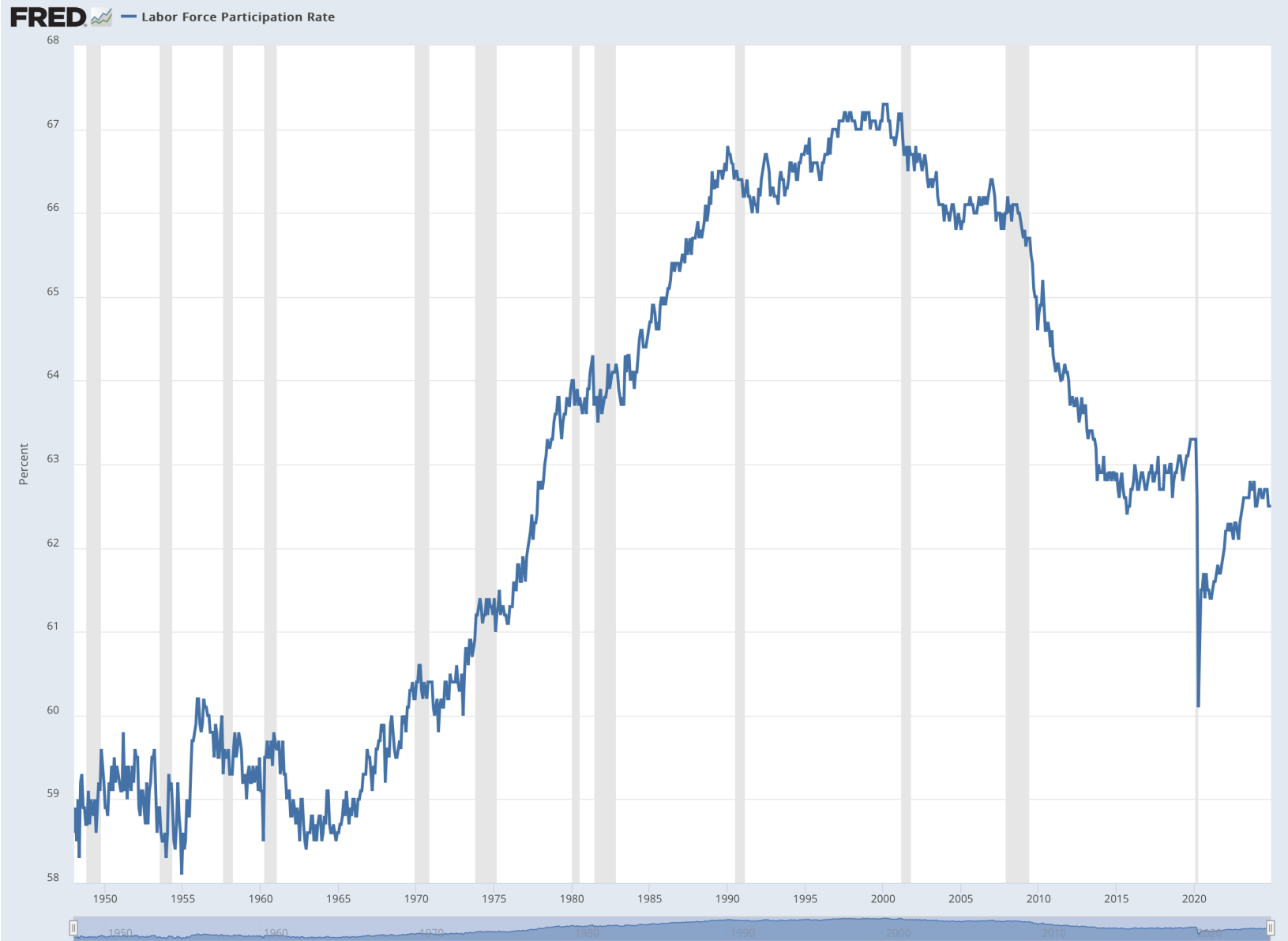


Data from the Conference Board on Tuesday showed consumer confidence in December slumped to its second-lowest level in five years

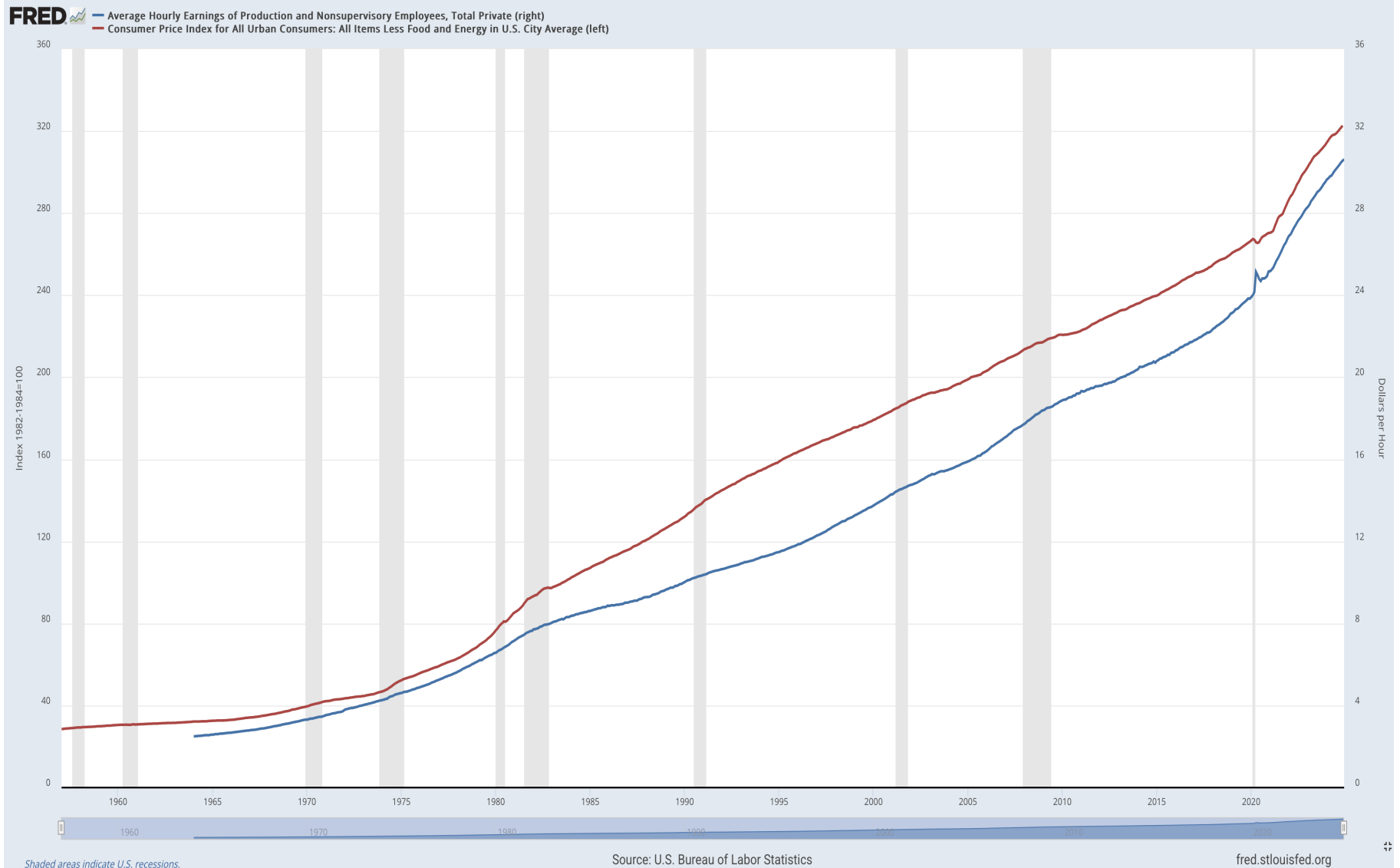
Unemployment Rate



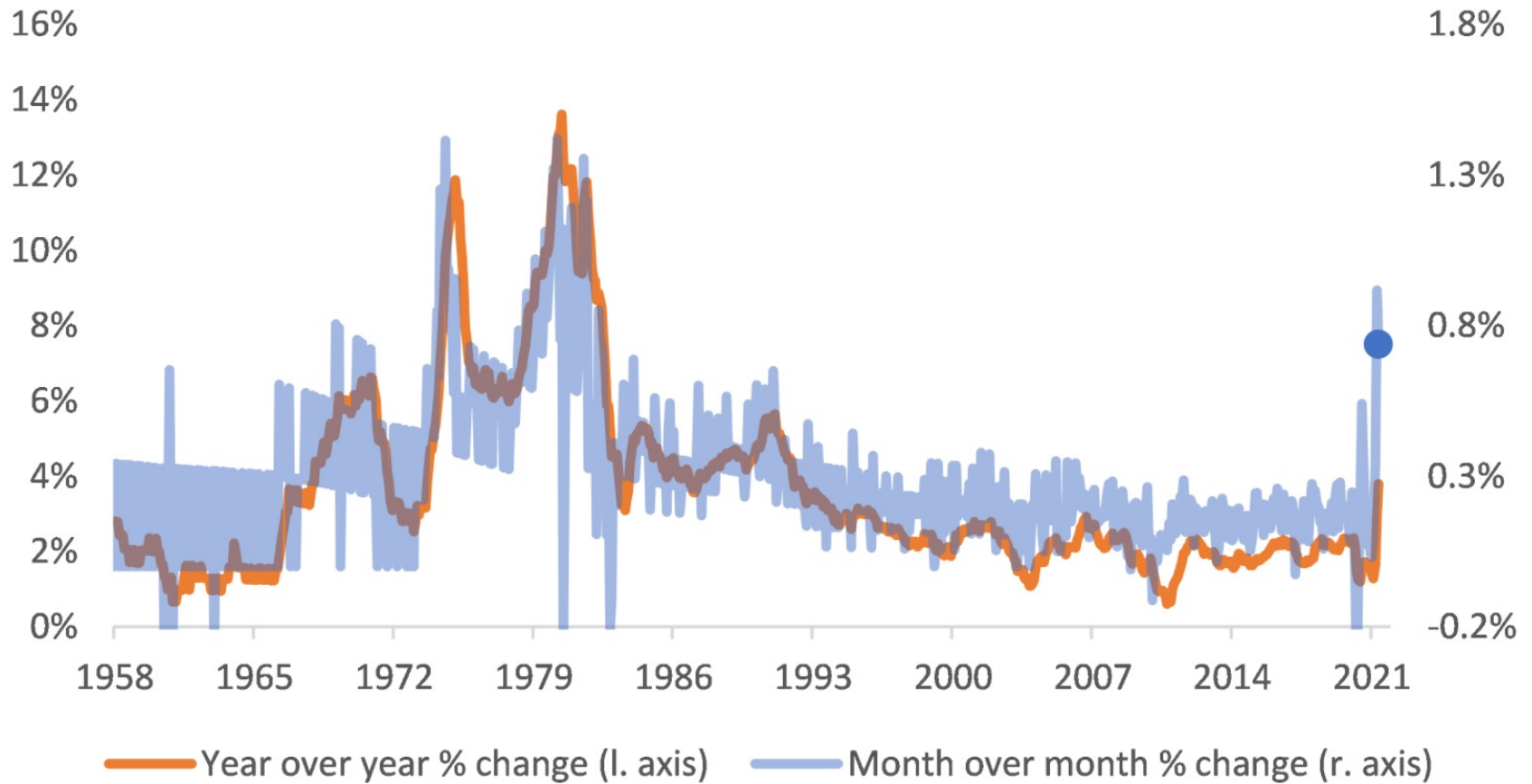
Labor Force Participation Rate

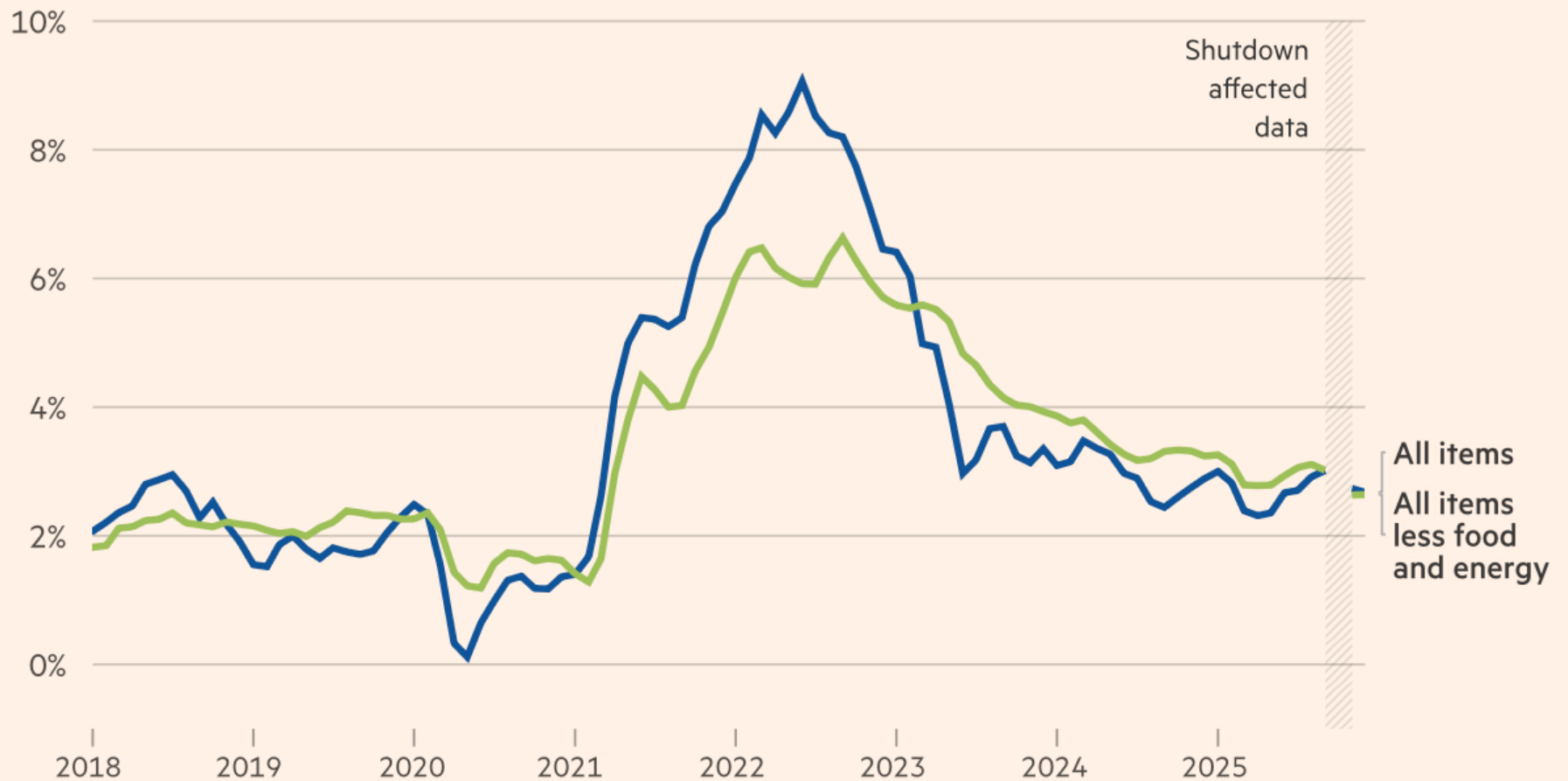


Wage Rate



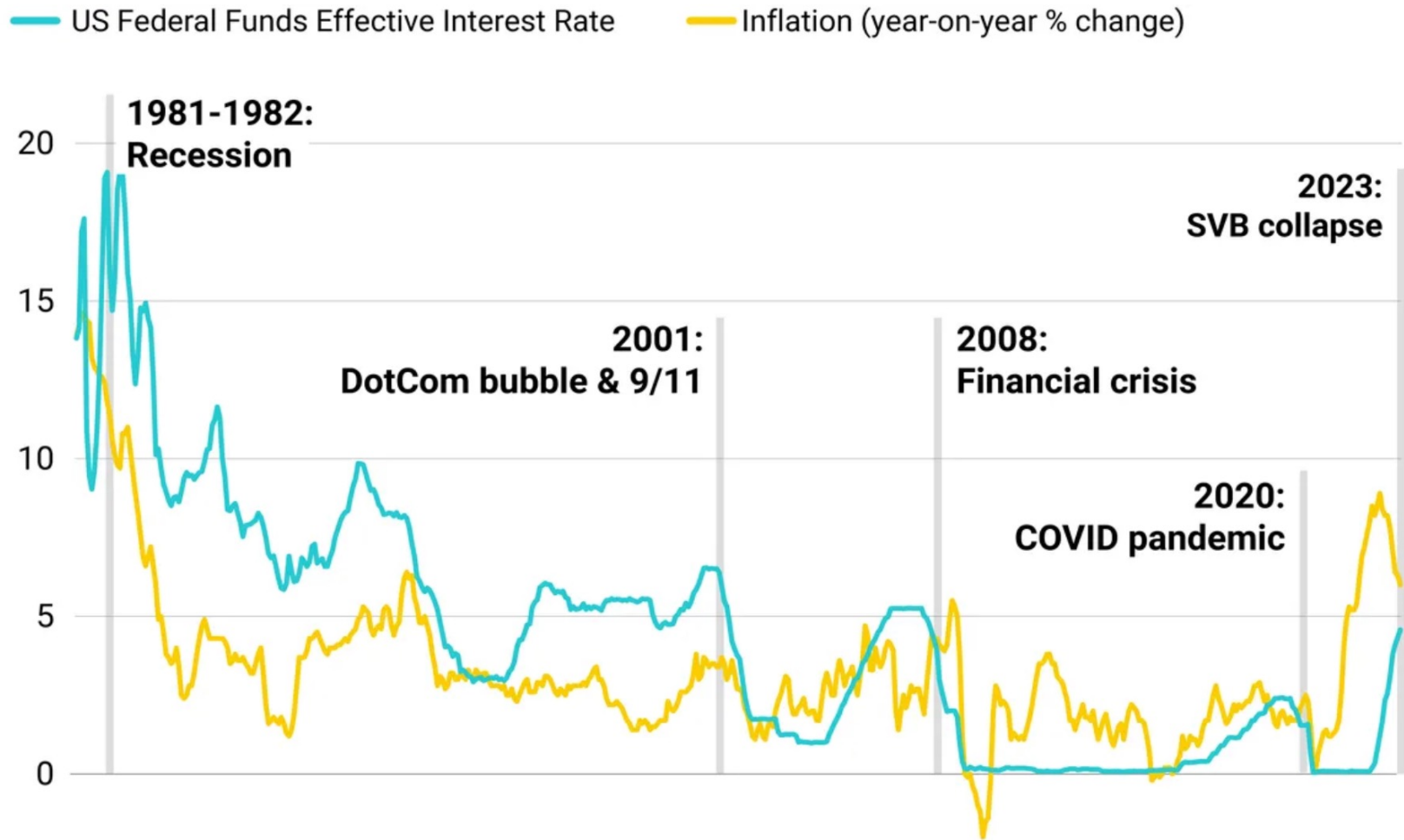
Core Inflation



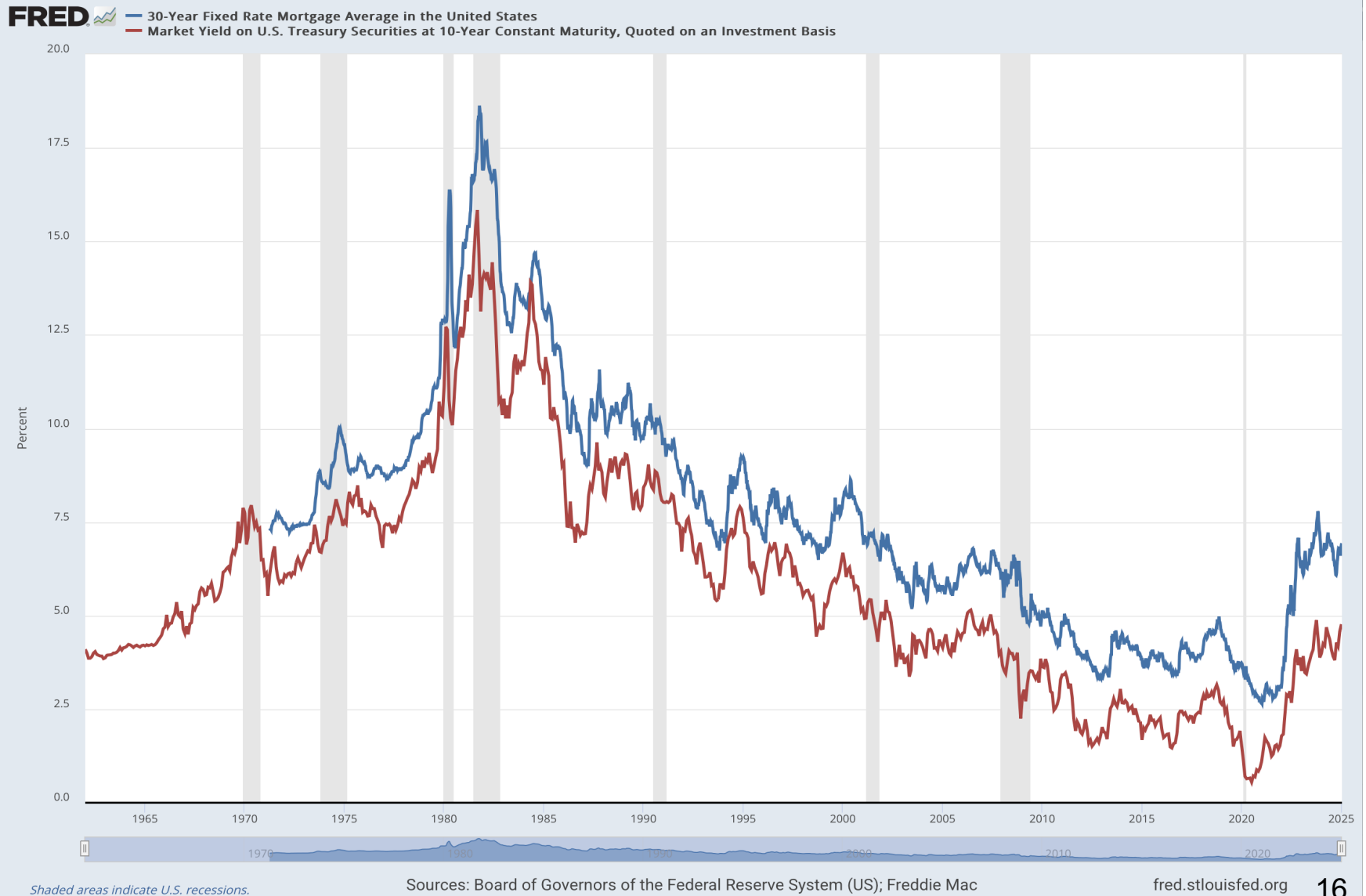


Source: [US Bureau of Labor Statistics](#) • Monthly changes are seasonally adjusted, yearly changes are unadjusted

Interest Rate

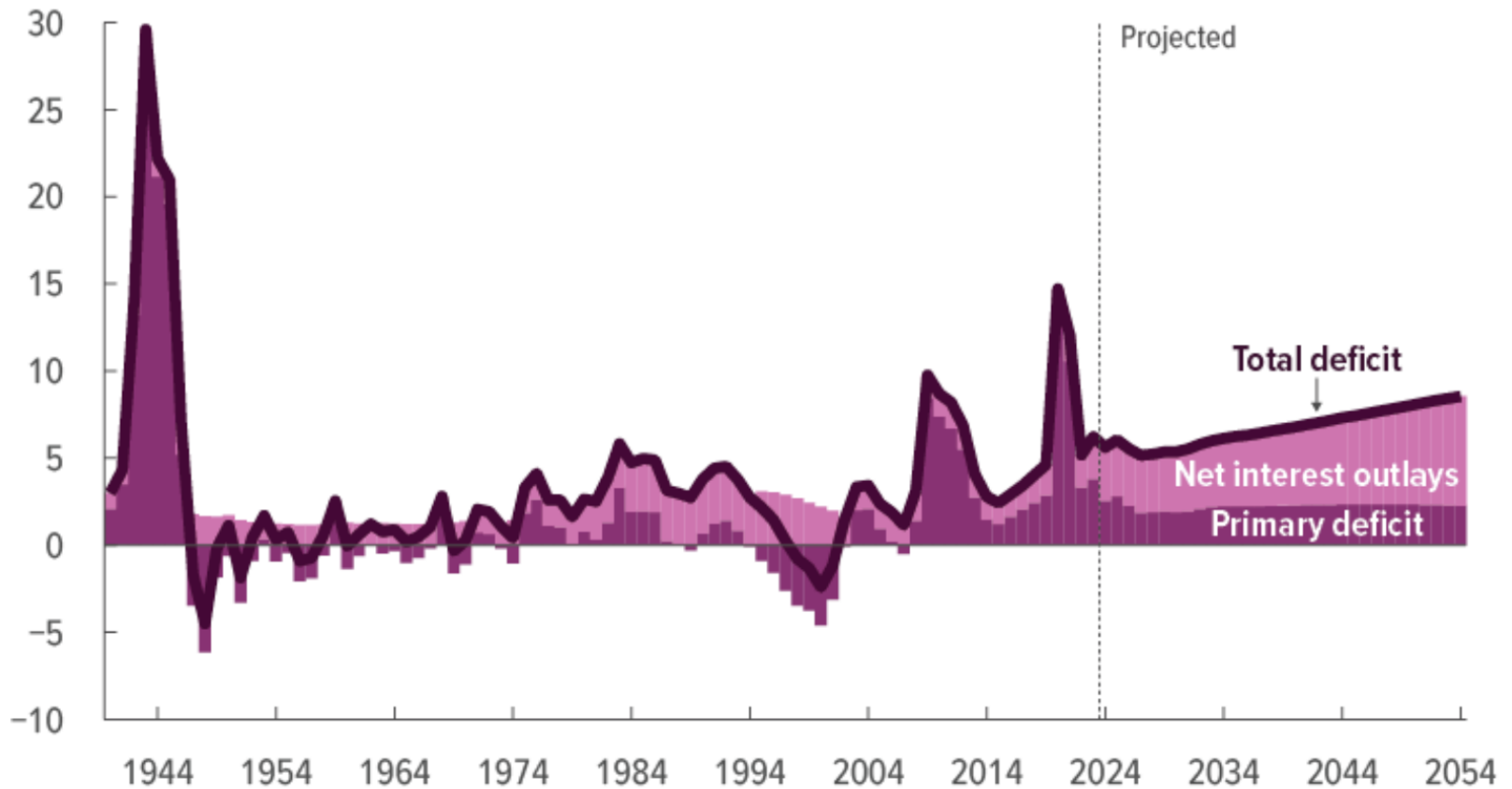


Mortgage Rate



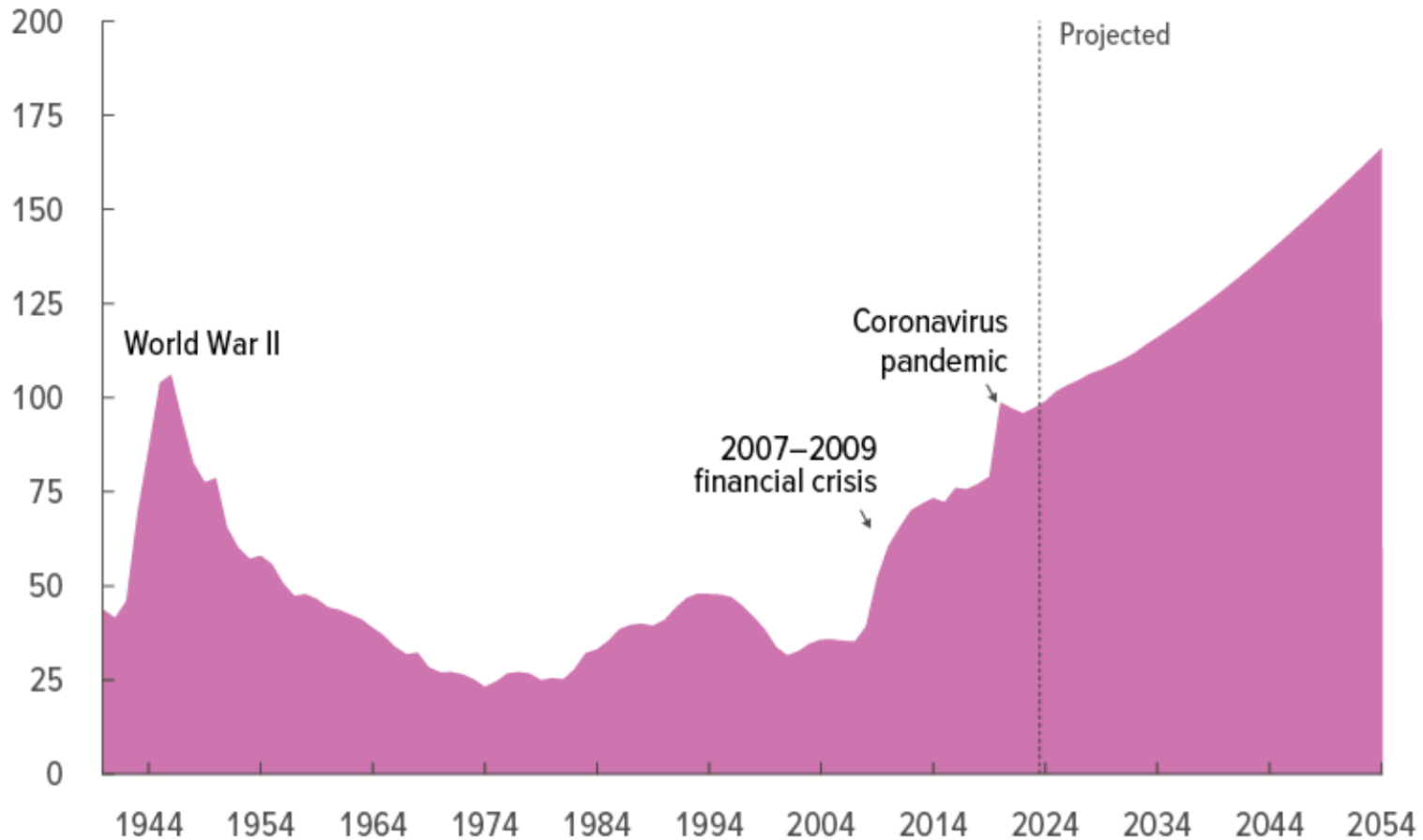
Government Deficit

Percentage of GDP

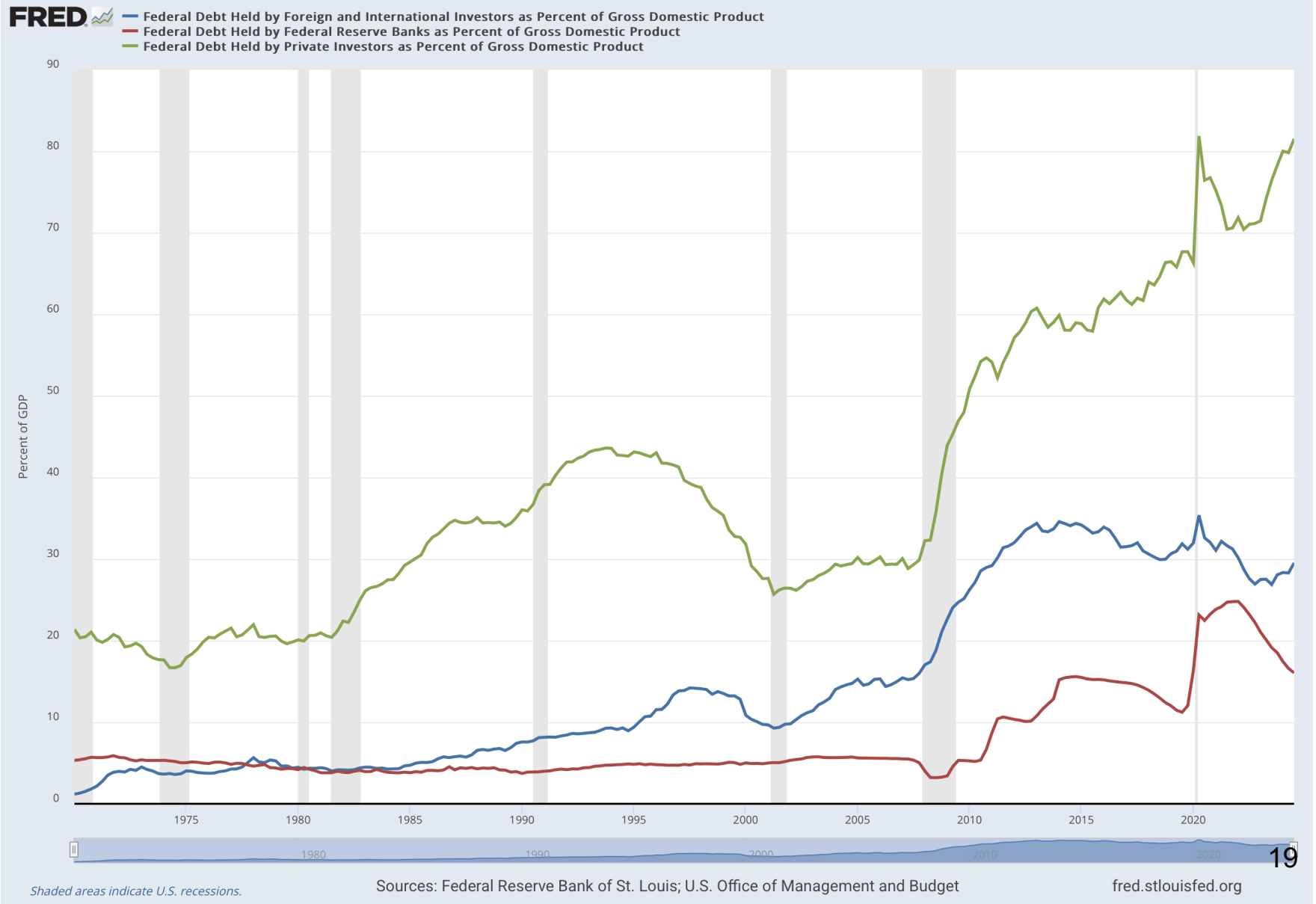


Public Debt

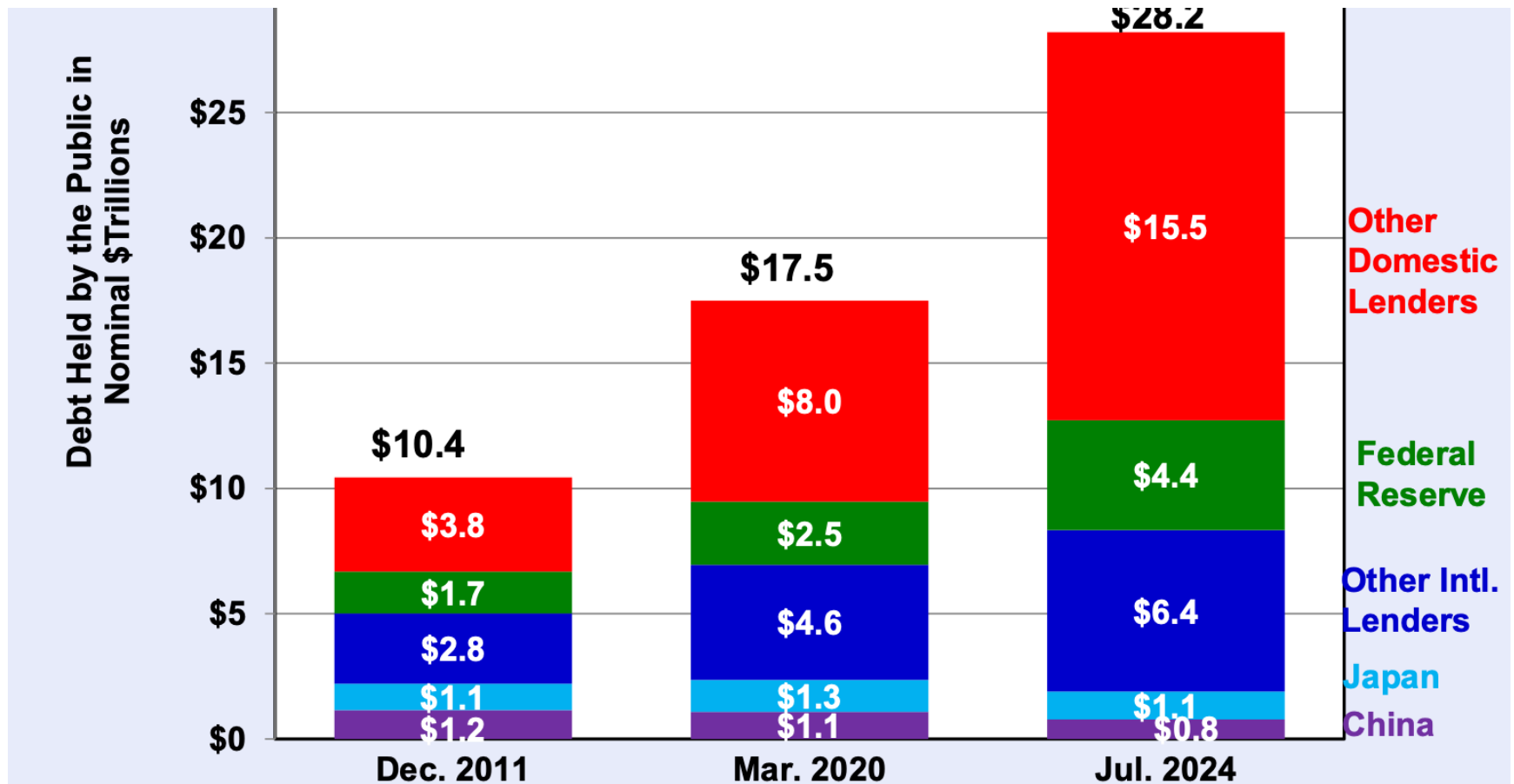
Percentage of GDP



Who Own US Debt

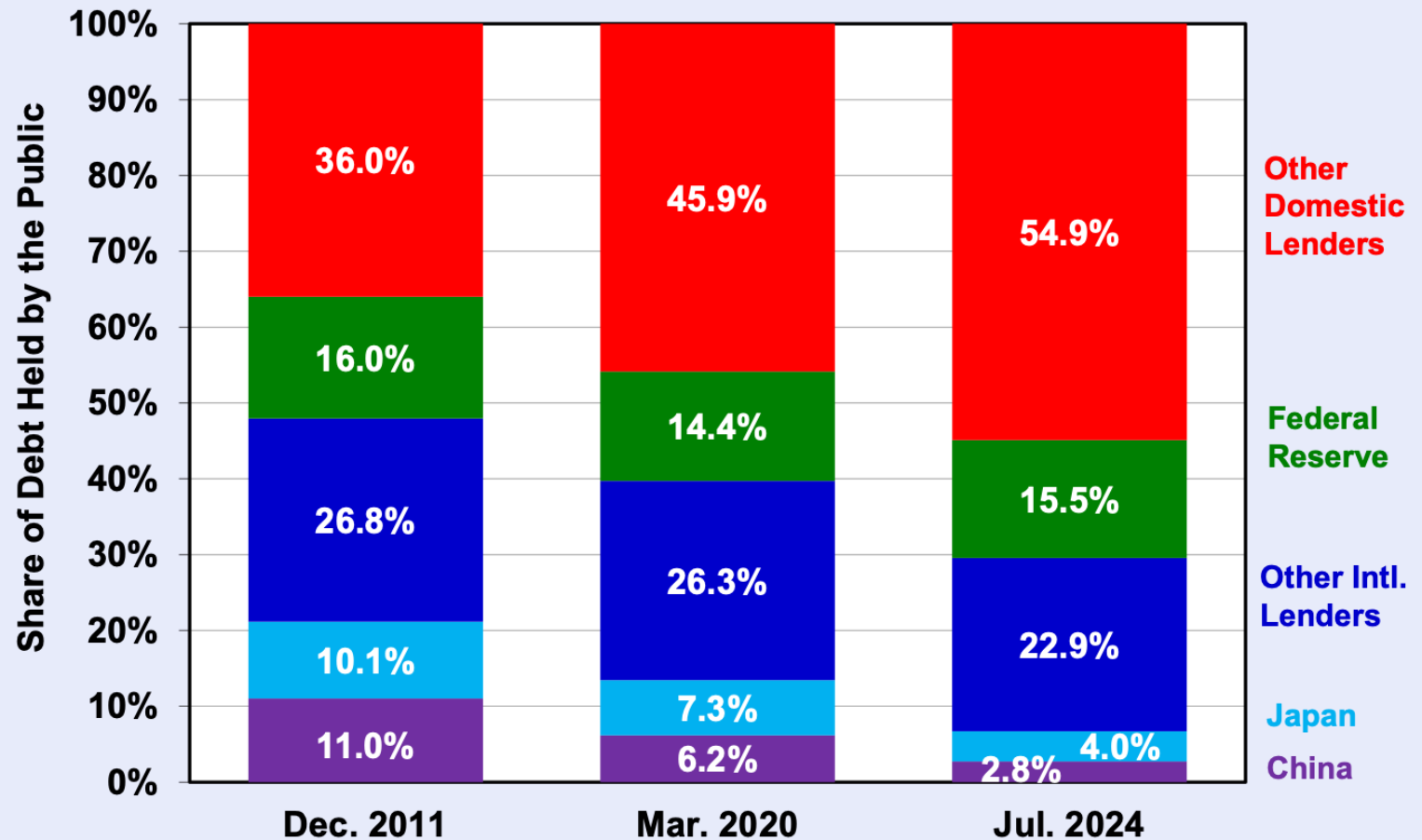


Who Own US Debt

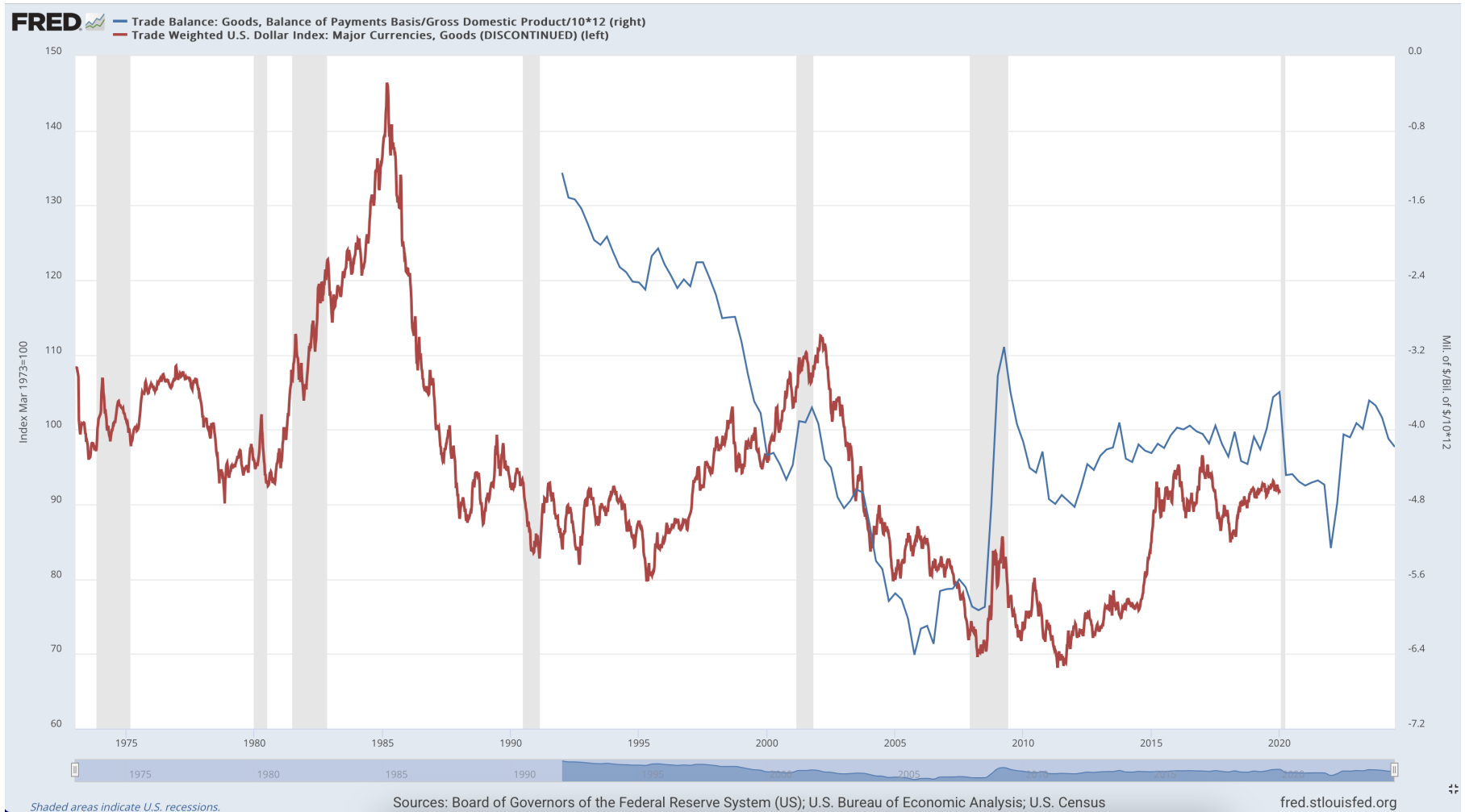


Source: U.S. Treasury, Federal Reserve. The Federal Reserve has expressed intentions to reduce its Treasury holdings.

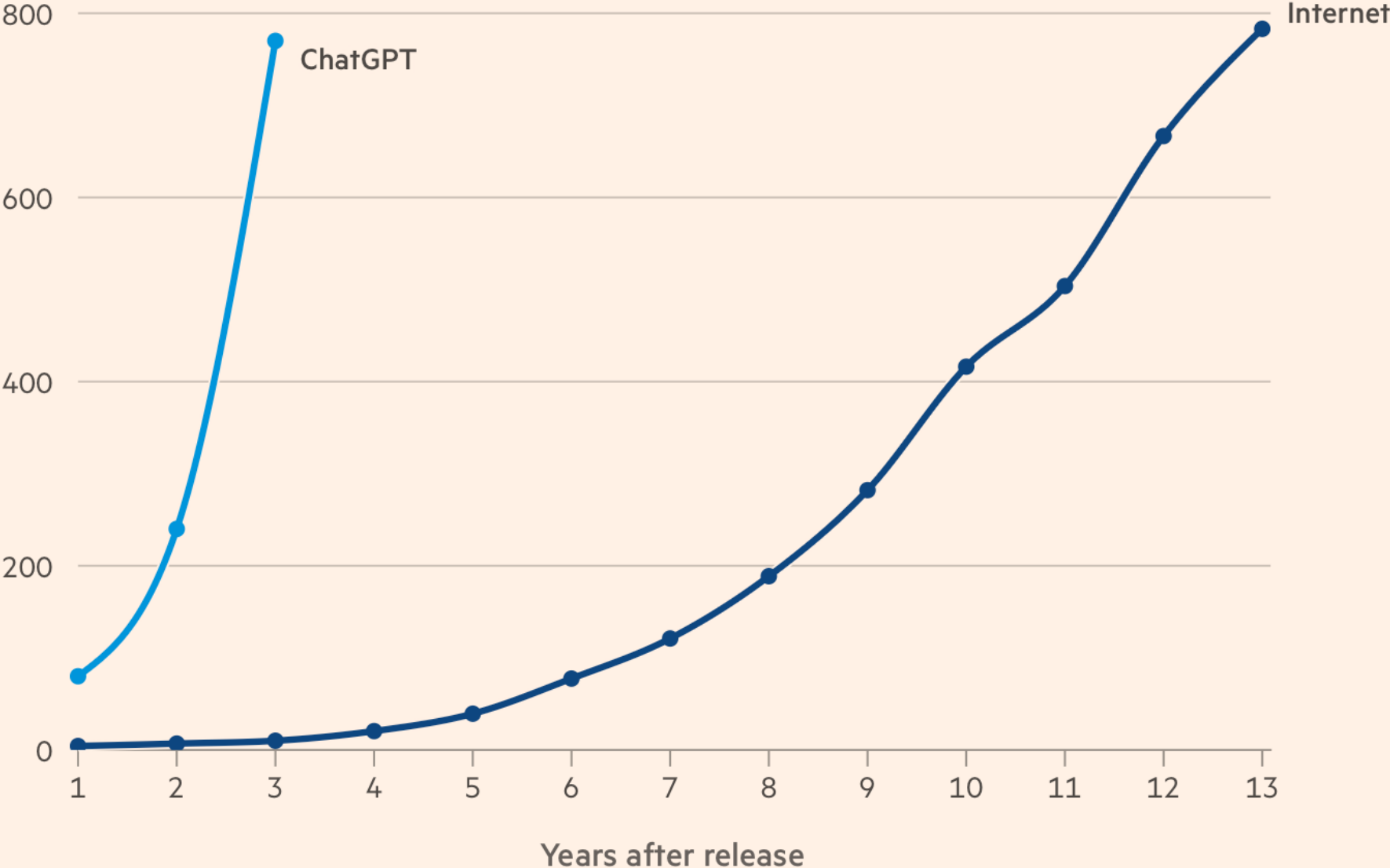
Who Own US Debt



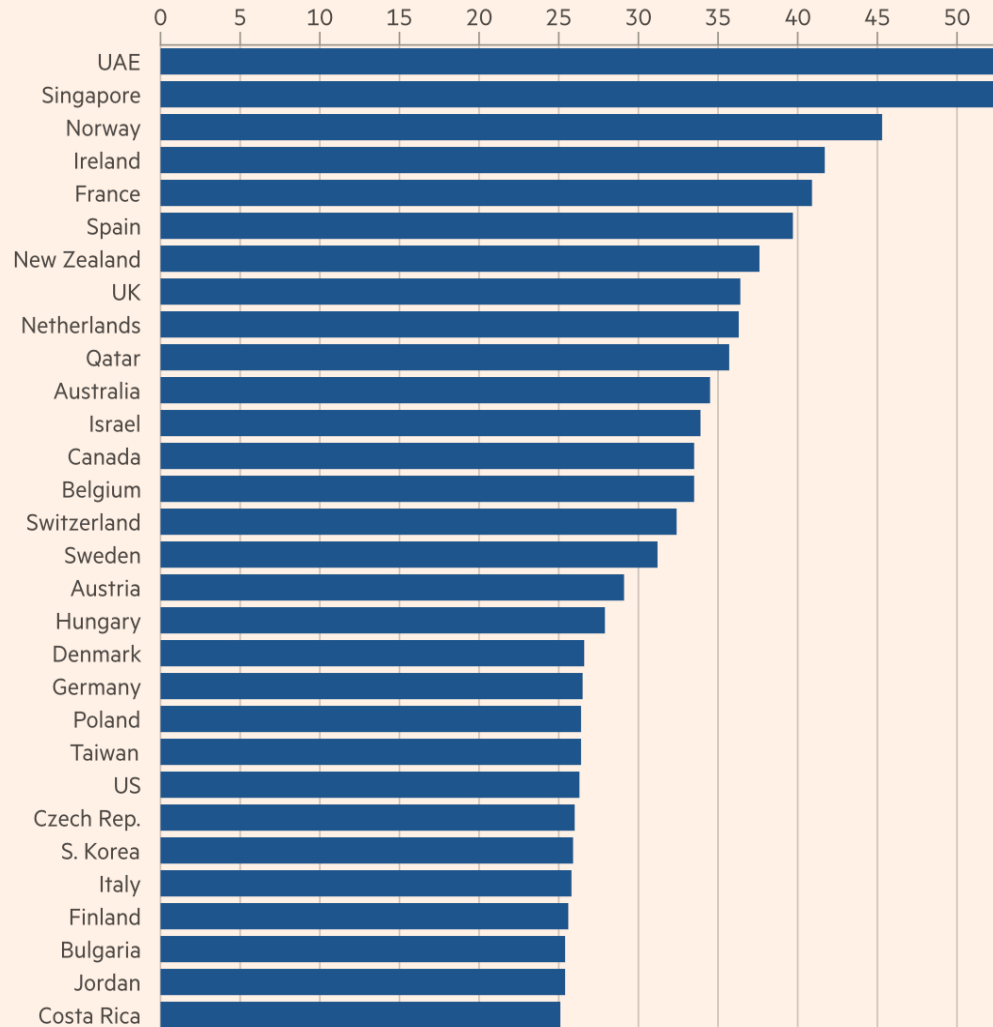
US Trade Balance



Users (mn)



The proportion of working-age population that uses AI, by country, %



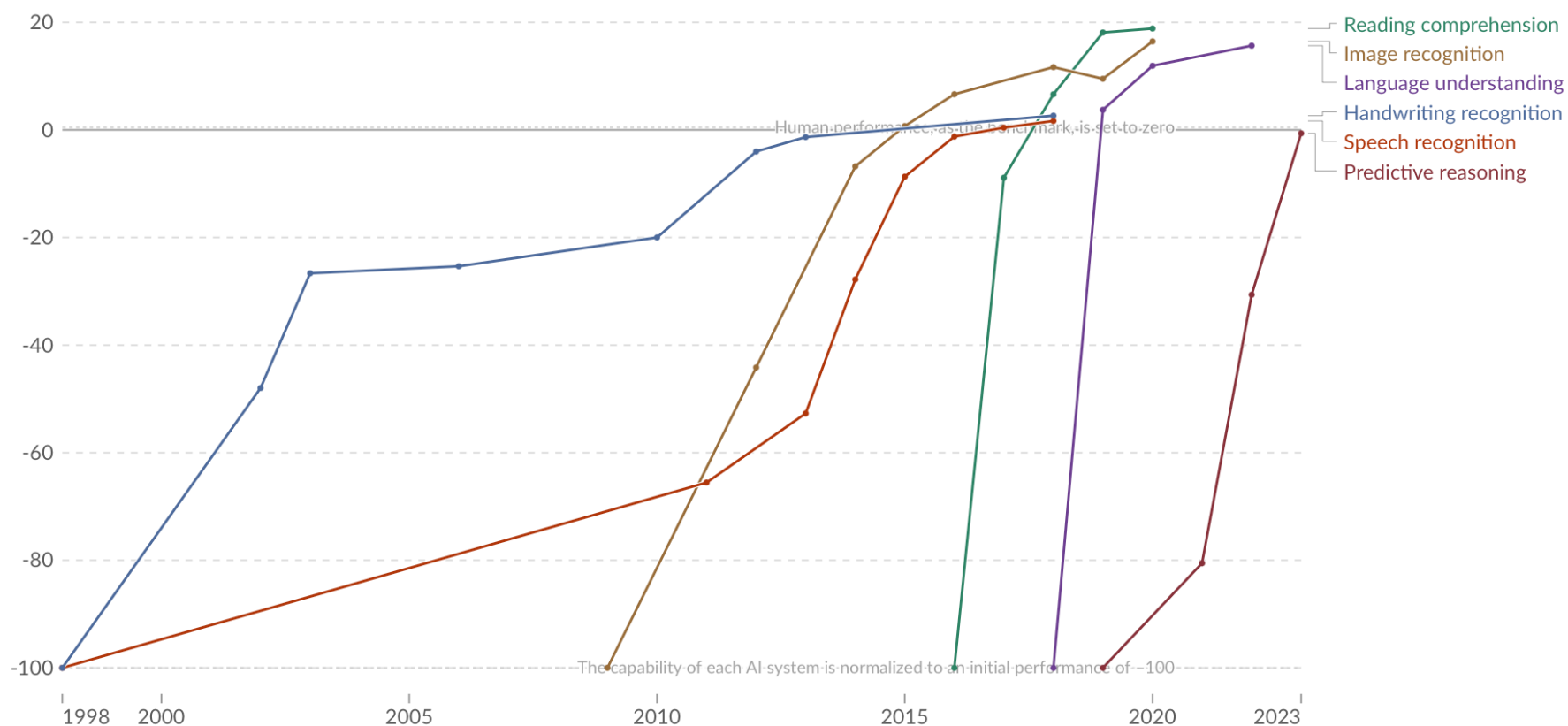
FINANCIAL TIMES

Source: AI for Good Lab - Mic

Test scores of AI systems on various capabilities relative to human performance

Within each domain, the initial performance of the AI is set to -100. Human performance is used as a baseline, set to zero. When the AI's performance crosses the zero line, it scored more points than humans.

Table Chart

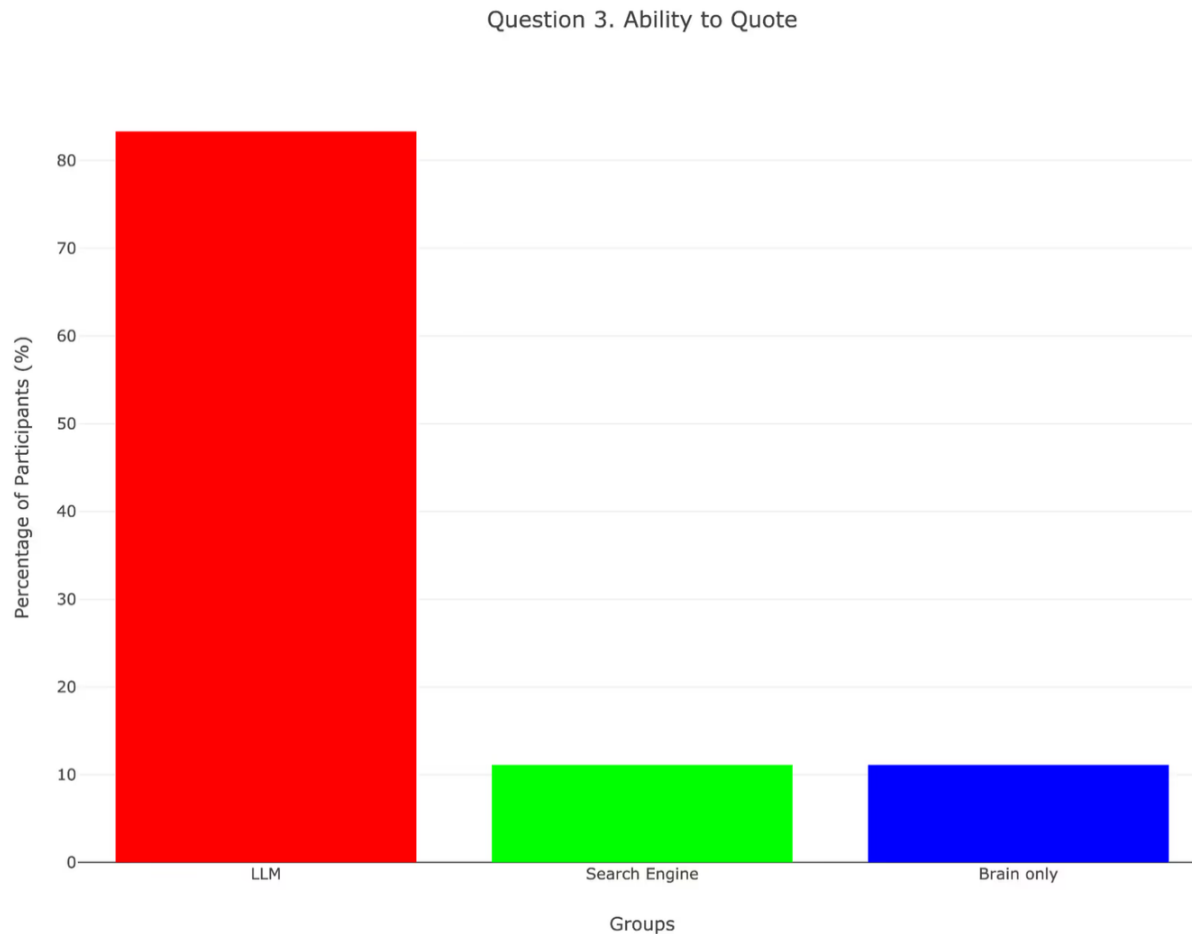


Data source: Kiela et al. (2023) – [Learn more about this data](#)

Note: For each capability, the first year always shows a baseline of -100, even if better performance was recorded later that year.

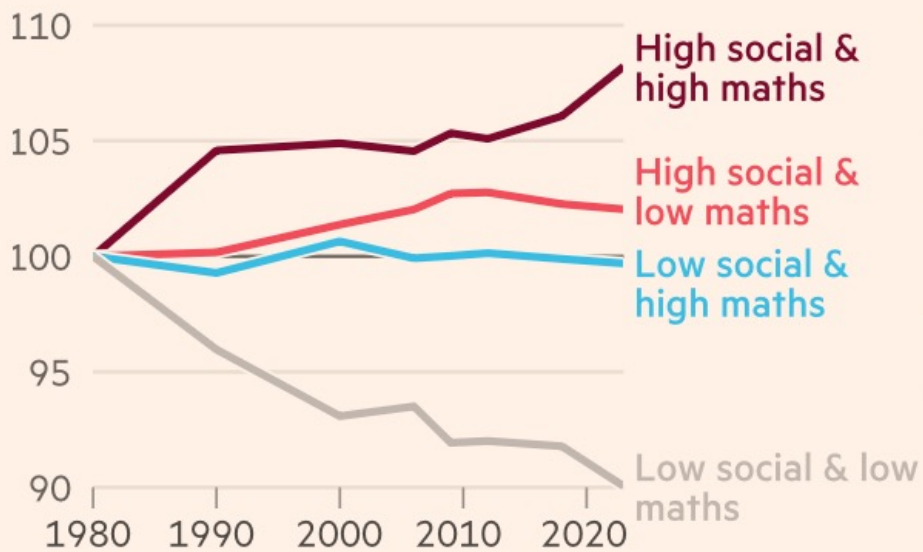
OurWorldinData.org/artificial-intelligence | CC BY

Figure 6. Percentage of participants within each group who struggled to quote anything from their essays in Session 1.

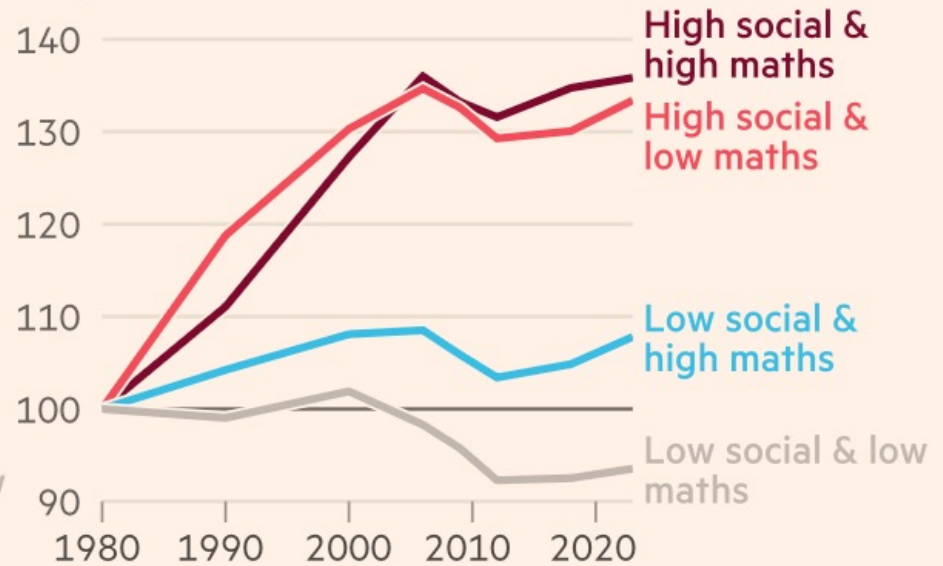


Trends in employment and real wages, split by relative importance of mathematical and social skills in the job (index, 1980 = 100)

Employment



Wages

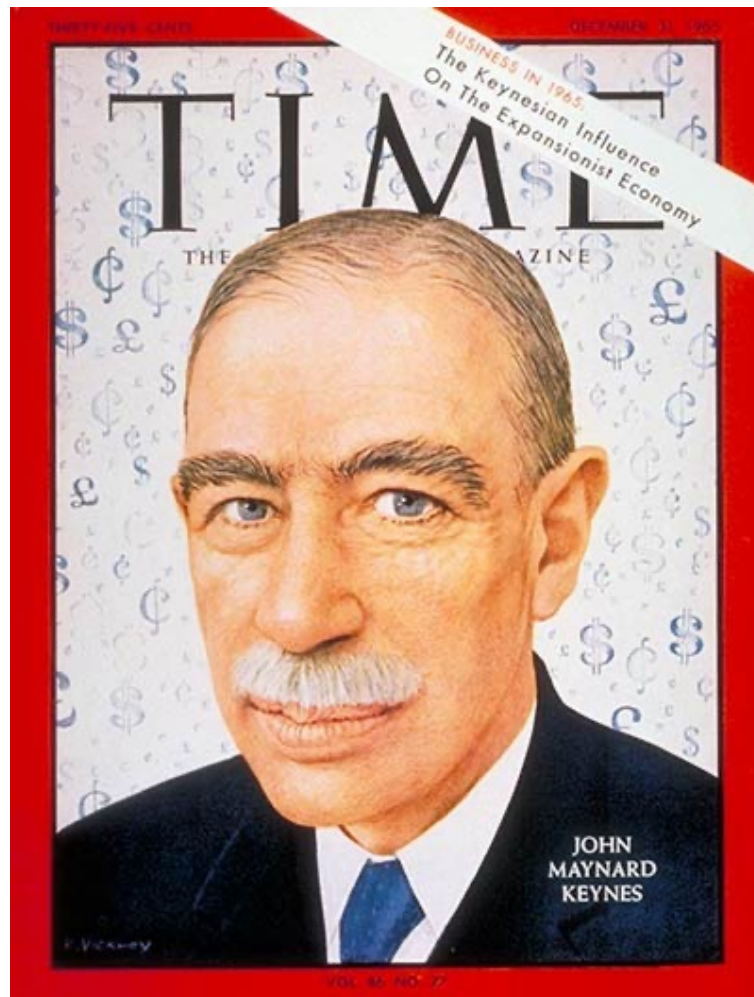


Sources: American Community Survey; O*Net; [The Growing Importance of Social Skills in the Labor Market \(Deming, 2017\)](#)

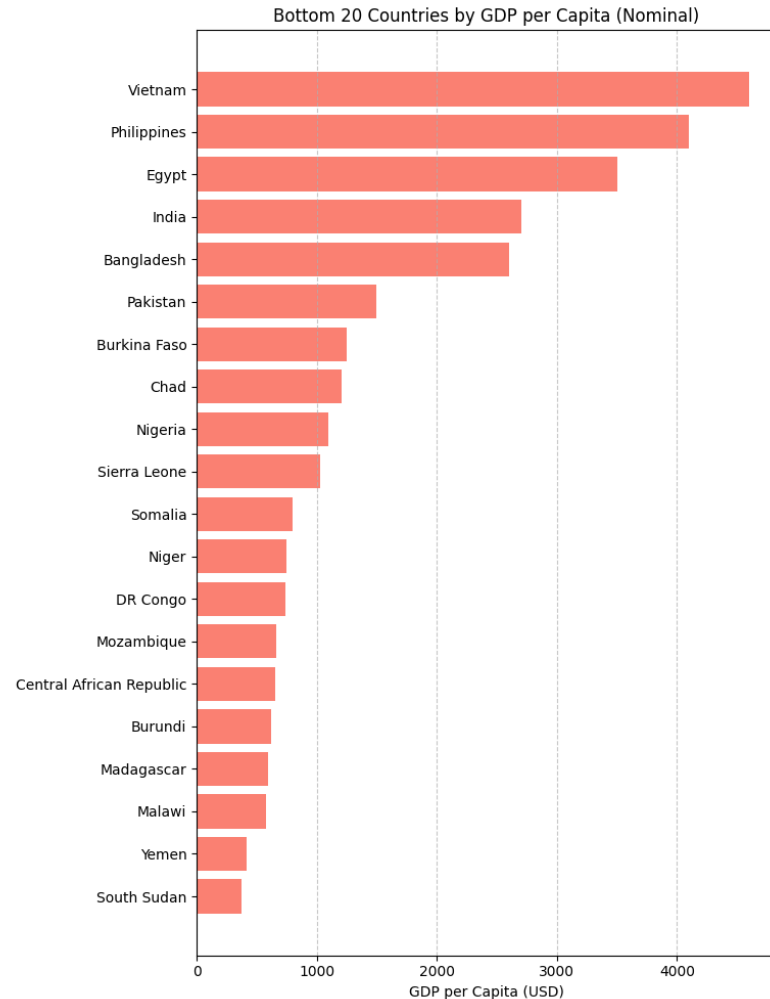
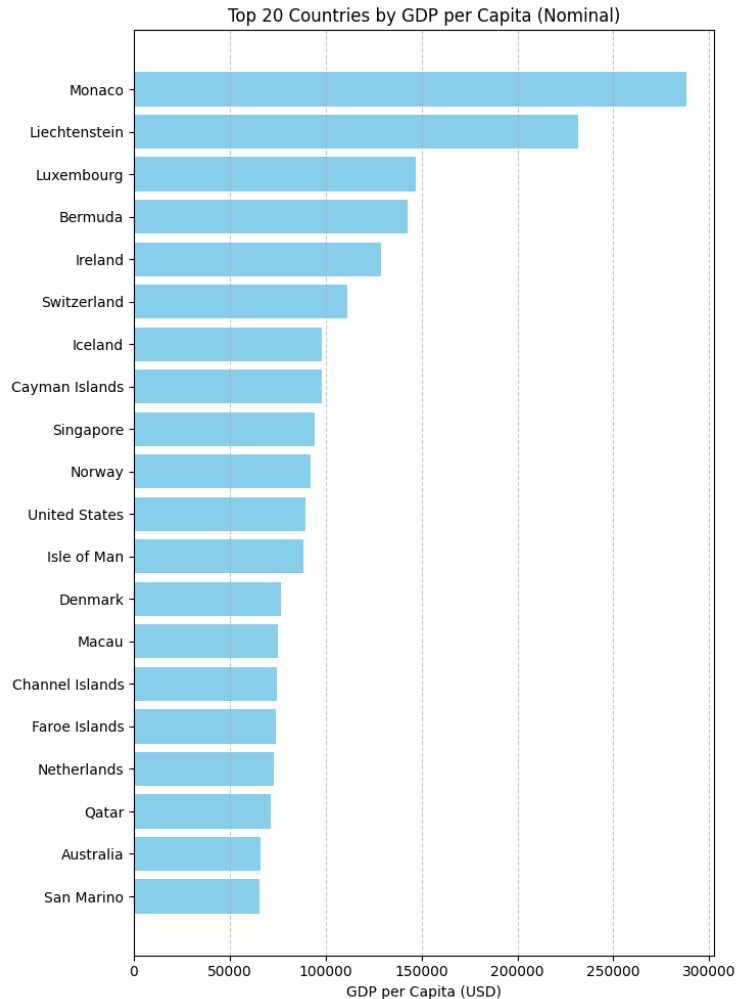
What is Macroeconomics

- Models built to explain macroeconomic phenomena.
- Examples of issues:
 - Why are so many countries poor?
 - Why are millions of people unemployed?
 - What makes people unequal in income and wealth?
 - Why are there recession? Can the government do anything to combat recession? Should it?
- To summarize: ***long-run growth*** and ***business cycles***.

Birth of modern macroeconomics



Why should we care about Macroeconomics?



Why should we care about Macroeconomics?



Why should we care about Macroeconomics?



What do Macroeconomists do?

- There are three types of macroeconomists:

1. **Forecasters:**

- predict how inflation, GDP or stock markets will perform in the short run. Unfortunately economists are pretty bad at predicting the short run, even worse than meteorologists.

2. **Analysts:**

- private sector: interpret current economic events and inform the decision of an investment bank to, consulting firms.
- public sector: inform the government on what are the right policies, given a certain political objective that wants to be reached.

3. **Researchers:**

- developing new models or validating or falsifying existing ones.

List of Topics

- Review of Some Key Concepts From Microeconomics
- National Income Accounting
- Labor market
- Macroeconomics: The Big Picture
- Economic Growth
- The Keynesian Model
- Macroeconomic Equilibrium
- Economic Policies